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**Paper Title: Social Context Modulates Speech Self-Monitoring:
Evidence from Asynchronous Voice Communication**

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Social Context Modulates Speech Self-Monitoring: Evidence from Asynchronous Voice Communication

1. Literature Review

1.1 Self-Monitoring in Speech Production

Speech production incorporates a self-monitoring system that detects and corrects errors during planning and articulation. The classical model posits that monitoring operates through an internal auditory feedback loop, wherein speakers compare ongoing speech output against intended representations (Levelt, 1989). This monitor exhibits a hierarchical temporal structure: monitoring activities are distributed across pre-articulatory planning and post-articulatory production stages, with distinct neural and behavioral signatures at each level (Dorokhova et al., 2024).

Behaviorally, self-monitoring manifests through multiple observables. Speech onset latency, the interval between a cue to speak and the initiation of vocalization, reflects the duration of pre-articulatory planning and the stringency of the monitoring gate applied before articulation (Levelt, 1989). Disfluencies, including filled pauses (e.g., "um," "uh"), repetitions, self-repairs, and silent pauses, index monitoring activity during and after articulation, marking moments when the system detects a mismatch and suspends output or initiates correction (Postma, 2000). Repairs occupy a distinctive position: they signal that an error has not only been detected but deemed sufficiently consequential to warrant explicit correction, making them a sensitive marker of elevated monitoring stringency.

1.2 Contextual Modulation of the Monitoring System

Recent neuroimaging evidence has established that the speech monitoring system is not static but exhibits pronounced contextual plasticity. Ozker and Hagoort (2025) demonstrated substantial individual variability in susceptibility to auditory feedback perturbations, with functional and structural correlates in right-lateralized

frontotemporal networks. Delayed auditory feedback paradigms have been shown to activate a right-lateralized speech monitoring network distinct from the left-dominant production network (Kort et al., 2014), indicating that the monitor can be dynamically upregulated in response to altered feedback conditions.

Existing paradigms, however, have overwhelmingly manipulated physical properties of the feedback signal (e.g., temporal delay, frequency shift), leaving unexamined whether social-contextual cues alone can modulate monitoring behavior (Meekings and Scott, 2025). In naturalistic communication, speakers routinely adjust production based on perceived audience characteristics, a phenomenon well-documented in sociolinguistics but largely absent from experimental investigations of monitoring. If monitoring is genuinely adaptive, it should respond to the anticipated communicative consequences of error, not only to degraded sensory feedback.

1.3 Asynchronous Voice Communication as a Social-Contextual Manipulation

Asynchronous voice communication, defined as vocal messaging in which production and reception are temporally decoupled (e.g., voice messages, social media voice notes), provides a theoretically motivated manipulation of social-contextual monitoring pressure. This communication mode is characterized by three features that collectively constitute an enhanced evaluative context. First, *delayed audience* eliminates the possibility of immediate interactive repair; errors cannot be corrected in real time through conversational grounding (Choi, 1996). Second, *message replayability* means that errors are not ephemeral but can be perceived multiple times, amplifying their communicative cost (Lind et al., 2014). Third, *indeterminate audience* introduces uncertainty about who will ultimately receive the message, complicating the speaker's model of listener tolerance for error (Brake, 2014).

These three features, following Oomen and Postma (2001), should tonically activate the monitoring system under sustained evaluative pressure, lowering the threshold for error detection and increasing both pre-articulatory caution (longer onset

latencies) and post-articulatory intervention (increased disfluency rates, particularly repairs).

1.4 Research Gap and Study Objectives

No prior study has tested whether social-contextual cues, instantiated without physical feedback manipulation, can modulate the behavioral expression of speech self-monitoring. The present study addresses this gap by comparing speech production under two conditions within a within-subjects design: a synchronous condition in which participants describe images to a co-present experimenter, and an asynchronous condition in which participants describe images while imagining an unseen, temporally displaced listener whose access to the recording is indefinite.

1.5 Research Questions and Hypotheses

Three research questions were formulated to examine distinct predictions of the monitoring-upregulation account at different temporal loci of production. The first concerns whether asynchronous voice context extends pre-articulatory planning: if the monitor is globally upregulated under evaluative pressure, speakers should require additional preparation before releasing the articulatory plan, yielding longer speech onset latencies in the asynchronous condition (H1: Onset Latency Hypothesis). The second addresses whether this upregulation produces a global disfluency increase: heightened monitoring sensitivity should lower the detection threshold throughout production, elevating the total frequency of all disfluency types combined (H2: Disfluency Hypothesis). The third targets the predicted behavioral signature of post-articulatory enhancement: because asynchronous context specifically amplifies the consequences of errors that have been produced and can be replayed, speakers should be more inclined to interrupt ongoing speech and issue explicit corrections, selectively increasing the proportion of repairs within the overall disfluency profile (H3: Repair Proportion Hypothesis).

2. Methods

2.1 Participants

Twenty-one adults (11 female; Mage = 22.3 years, SD = 3.1) participated in the experiment. Sixteen participants were native Mandarin Chinese speakers; five were native English speakers. All participants reported normal or corrected-to-normal vision and no history of speech, language, or hearing impairment. Participants provided informed consent prior to the experiment and were debriefed following completion.

2.2 Stimuli and Task

Fifteen full-color drawings depicting everyday scenes and object arrays served as experimental stimuli. Each image was standardized to 1024×768 pixel resolution and presented at the center of the display against a white background. An additional four images were used exclusively during the practice phase. All images were selected to elicit spontaneous, multi-utterance descriptions without requiring specialized vocabulary.

The experiment was implemented in PsychoPy (Peirce et al., 2019) and administered individually in a quiet laboratory setting. For the synchronous condition, participants were informed that they would describe images directly to the experimenter, who would be present and attending. For the asynchronous condition, participants were informed that their descriptions would be recorded and later played back to an unidentified listener located behind a screen; the experimenter then positioned themselves away from the participant's line of sight. The experimenter verbally confirmed comprehension before proceeding.

Participants completed four practice trials (two per condition) using images distinct from the experimental set. The experiment then proceeded through two blocks counterbalanced between two conditions, each corresponding to one condition. Each block contained 15 trials in which all 15 experimental images were presented in randomized order.

Each trial followed a fixed temporal sequence. (a) The target image was displayed centrally for 5 s, during which participants viewed the image silently and prepared their description. (b) A response cue was then superimposed: the text "Go!" (synchronous condition) or a recording icon (asynchronous condition), signaling the onset of the 15 s speaking phase, during which participants produced a spontaneous, unstructured description. (c) At the end of the speaking phase, a visual countdown ("3," "2," "1") was displayed at 1 s intervals, providing a predictable window to conclude the description. (d) A fixation cross was presented for 1.5 s, demarcating the inter-trial interval. The total trial duration was approximately 21.5 s. Audio was recorded continuously at 48 kHz monaural within each block using a PsychoPy-integrated PortAudio microphone backend; the microphone was polled for data integrity between trials using a non-blocking routine with automatic fragment merging when the internal buffer exceeded three audio chunks.

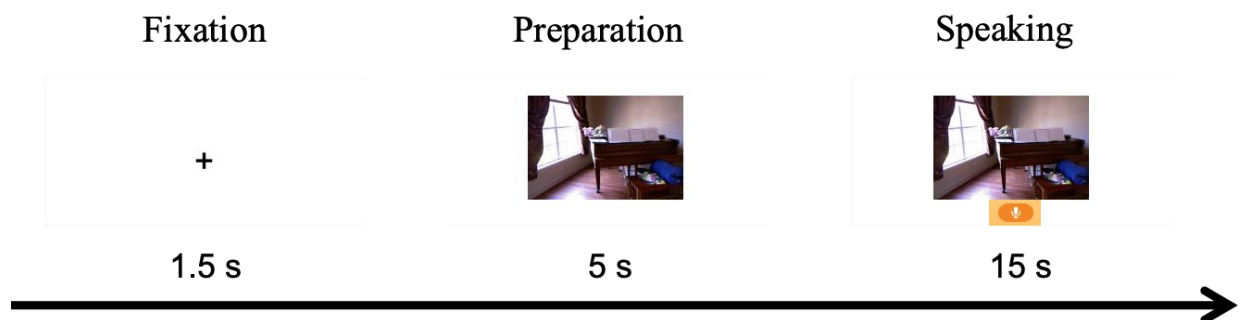


Figure 1 Schematic of trial structure with timing parameters

2.3 Annotation and Coding

The continuous block-level audio recordings were segmented and annotated using a custom Praat-based pipeline. A seven-tier TextGrid was generated programmatically for each participant and test block using Parselmouth (Jadoul et al., 2018) for acoustic analysis.

Silent pauses and speech onsets were detected automatically using adaptive intensity thresholding computed per trial. For each trial's speech window, two

thresholds were derived from the intensity distribution: the silence threshold at the 20th percentile ($P_{10} + 0.25 \times (P_{90} - P_{10})$) and the speech onset threshold at the 35th percentile ($P_{10} + 0.35 \times (P_{90} - P_{10})$). Silent pauses were intervals during which intensity remained below the silence threshold for a minimum of 150 ms. Adjacent pauses separated by less than 100 ms of intervening speech were merged to avoid fragmenting perceptually unitary pauses. A speech offset detection routine identified the last point of sustained suprathreshold intensity within each trial window, and silent intervals occurring entirely after this point were discarded as trailing silence.

Speech onset latency was detected by scanning forward from the go-signal onset; the first point at which intensity remained continuously above the onset threshold for at least 50 ms was marked as speech onset, and latency was computed as the difference between this timestamp and the go-signal onset, in milliseconds.

The remaining disfluency categories were manually annotated. Trained annotators following a standardized protocol labeled intervals on three dedicated tiers within Praat. Filled pauses were defined as non-lexical vocalizations (e.g., "um," "uh," "嗯," "那个"). Repetitions were defined as verbatim or near-verbatim reproduction of a word or phrase without syntactic or semantic modification. Repairs were defined as instances in which the speaker interrupted ongoing speech to correct a phonological, lexical, or syntactic error. Annotators used waveform and spectrogram displays to identify interval boundaries; a fourth tier for orthographic transcription was provided but not required for the quantitative analyses. The total disfluencies is the sum of all four types.

3. Results

3.1 Confirmatory Hypothesis Tests

The three hypotheses were evaluated in sequence, moving from global predictions about monitoring upregulation to a specific prediction about post-articulatory correction. For onset latency (H1), the mean in the asynchronous condition ($M = 1196.98$ ms, $SD = 379.14$) exceeded that in the synchronous condition

($M = 1157.09$ ms, $SD = 264.44$) by 39.89 ms, a difference that did not approach statistical significance, $t(20) = 0.691$, $p = .249$, $d = 0.151$. For total disfluencies (H2), the asynchronous mean ($M = 6.21$, $SD = 2.99$) was marginally lower than the synchronous mean ($M = 6.31$, $SD = 3.37$), a difference of -0.10 opposite in direction to the predicted effect, $t(20) = -0.425$, $p = .662$, $d = -0.093$. For repair proportion (H3), however, the asynchronous condition ($M = 0.064$, $SD = 0.064$) yielded a significantly higher value than the synchronous condition ($M = 0.040$, $SD = 0.046$), $t(20) = 1.885$, $p = .037$ (one-tailed), $d = 0.412$. H1 and H2 were not supported; H3 was supported.

The divergent outcomes for H2 and H3, a null total-disfluency effect coupled with a significant repair-proportion effect, indicated that the asynchronous manipulation did not exert a uniform influence across disfluency categories. We therefore decomposed the disfluency data by type to test whether the asynchronous condition's effect was specific to particular disfluency categories.

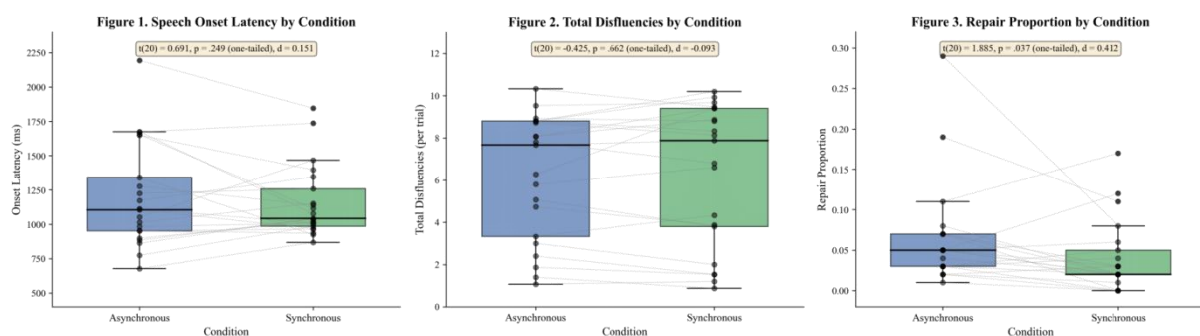


Figure 2 Panel of bar plots for onset latency, total disfluencies, and repair proportion by condition

Table 1. Paired-Samples t-Test Results for Main Analyses

Dependent Variable	Mean Difference (Asynchronous - Synchronous)	t	p	Cohen's d
Onset Latency (ms)	39.89	0.691	0.249	0.151
Total Disfluencies	-0.10	-0.425	0.662	-0.093
Repair Proportion	0.024	1.885	.037*	0.412

Note. $N = 21$. * $p < .05$ (one-tailed). Cohen's d = mean difference divided by standard deviation of the differences.

3.2 Condition × Disfluency Type Interaction

To determine whether the asynchronous condition's effect was type-specific, a 2 (Condition: synchronous vs. asynchronous) × 4 (Disfluency Type: filled pauses, repetitions, repairs, silent pauses) repeated-measures analysis of variance was conducted on disfluency frequencies. Mauchly's test indicated violations of sphericity for Type ($p < .001$) and the Condition × Type interaction ($p < .001$); Greenhouse-Geisser corrected values are reported throughout.

The main effect of Condition was significant, $F(1, 20) = 30.054$, $p < .001$, $\eta^2 = 0.600$, as was the main effect of Type, $F(3, 60) = 24.544$, $p < .001$, $\eta^2 = 0.551$. Critically, the Condition × Type interaction was significant and large, $F(3, 60) = 75.328$, $p < .001$, $\eta^2 = 0.790$, confirming that the effect of asynchronous context was not uniform across disfluency types.

Post-hoc paired-samples t-tests comparing conditions for each disfluency type were conducted with a Bonferroni correction for four comparisons (adjusted $\alpha = .0125$). Only repairs exhibited a significant condition difference: repair frequency was higher in the asynchronous condition ($M = 0.28$, $SD = 0.12$) than in the synchronous condition ($M = 0.18$, $SD = 0.16$), $t(20) = 2.684$, $p = .007$, $d = 0.58$, a medium-sized effect. Filled pauses (asynchronous: $M = 2.04$, $SD = 1.45$; synchronous: $M = 2.27$, $SD = 1.63$; $p = .964$, $d = -0.42$), repetitions (asynchronous: $M = 0.26$, $SD = 0.24$; synchronous: $M = 0.22$, $SD = 0.31$; $p = .221$, $d = 0.17$), and silent pauses (asynchronous: $M = 3.63$, $SD = 1.58$; synchronous: $M = 3.64$, $SD = 1.77$; $p = .971$, $d = -0.02$) did not differ between conditions. Thus, the significant repair proportion effect (H3) was driven exclusively by an elevation in repair frequency, not by a compensatory reduction in other disfluency types; the null total-disfluency result (H2) reflected the fact that the increase in repairs was offset by small, nonsignificant decreases in filled pauses and silent pauses.

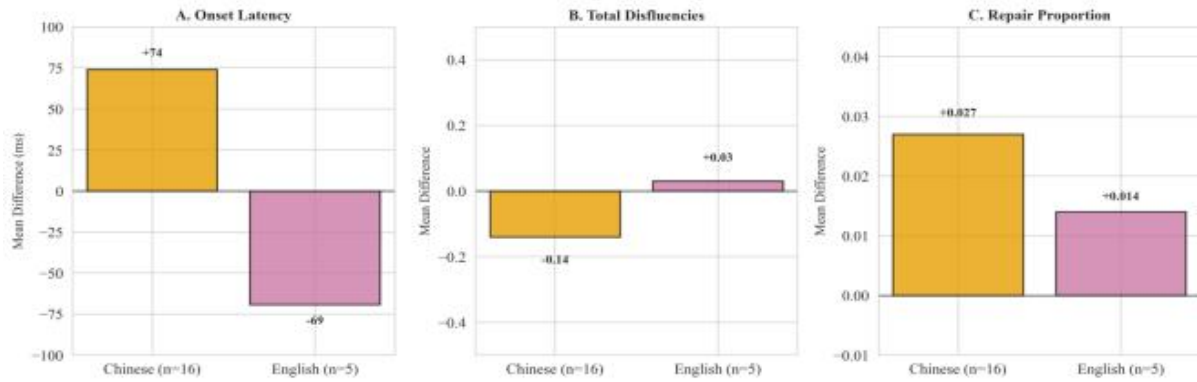


Figure 3: Interaction plot showing disfluency frequency by condition and type

Table 2. Repeated-Measures ANOVA Results for Disfluency Frequencies

Source	SS	df	MS	F	p	η^2
Condition	22.788	1	22.788	30.054	< .001***	0.6
Error (Condition)	15.165	20	0.758			
Type	23.622	3	7.874	24.544	< .001***	0.551
Error (Type)	19.249	60	0.321			
Condition $\times$ Type	297.743	3	99.248	75.328	< .001***	0.79
Error (Condition $\times$ Type)	79.052	60	1.318			

Note. $N = 21$. η^2 = partial eta squared. Mauchly's test indicated violation of sphericity for Type ($p < .001$) and Condition $\times$ Type ($p < .001$); Greenhouse-Geisser corrected values are reported.

3.3 Exploratory Analysis: Cross-Language Descriptive Comparison

To assess whether the observed effects exhibited cross-language consistency, descriptive statistics were computed separately for the Chinese L1 subgroup ($n = 16$) and the English L1 subgroup ($n = 5$). Formal inferential comparisons were precluded by the small and unequal sample sizes and the confound between language and experimental setting. Descriptively, the repair proportion advantage for the asynchronous condition was directionally consistent across groups (Chinese: $\Delta M = 0.027$; English: $\Delta M = 0.014$). Onset latency patterns were directionally inconsistent (Chinese: $\Delta M = +73.98$ ms; English: $\Delta M = -69.21$ ms), and total disfluency differences were near zero for both groups.

4. Discussion

The present study tested whether asynchronous voice context, a purely social-contextual manipulation, is sufficient to modulate the behavioral expression of speech self-monitoring. The findings reveal a selective rather than global modulatory effect: asynchronous context did not significantly alter speech onset latency or overall disfluency rates, but it did significantly increase the proportion of self-repairs, an effect driven by a specific elevation in repair frequency ($d = 0.58$) confirmed through a significant Condition $\times$ Disfluency Type interaction ($\eta^2 = 0.79$).

4.1 Selective Upregulation of Post-Articulatory Monitoring

The dissociation between onset latency (null) and repair proportion (significant) carries theoretical implications for the architecture of speech monitoring. Onset latency is conventionally interpreted as an index of pre-articulatory monitoring: the time required for the planning system to reach a confidence threshold sufficient to release the articulatory plan (Levelt, 1989). The null onset latency effect suggests that asynchronous voice context does not substantially alter the stringency of this pre-articulatory gate. One methodological factor warrants consideration: the 5 s silent viewing period preceding the speaking cue may have allowed participants to complete the bulk of their linguistic planning before the monitoring manipulation became operative, potentially attenuating condition differences on this measure.

In contrast, the significant elevation of repair proportion, and the specificity of the Condition $\times$ Type interaction to repairs alone, indicates that asynchronous context selectively upregulates post-articulatory monitoring. Repairs are the behavioral signature of a monitor that has not only detected a discrepancy but has judged it sufficiently consequential to interrupt ongoing speech and issue a correction. This pattern fits the characterization of asynchronous voice communication: the features of delayed audience, replayability, and indeterminate listenership primarily amplify the consequences of errors that have already been produced, rather than the probability of errors during planning. A speaker who knows their message may be replayed and

scrutinized by an unknown audience has a strong incentive to correct demonstrable errors, but this knowledge may not substantially alter the initial planning process, which is constrained more by lexical access and syntactic encoding than by social-evaluative considerations.

4.2 Interaction Profile and the Specificity of the Effect

The Condition $\times$ Disfluency Type interaction provides fine-grained insight into which monitoring sub-processes are socially sensitive. Filled pauses, often interpreted as signals of lexical retrieval difficulty or planning uncertainty (Clark and Fox Tree, 2002), were not modulated by condition. Silent pauses, which index planning boundaries and retrieval difficulty, were likewise unaffected. Repetitions, which may reflect a lower-cost floor-holding strategy during planning delays, showed no condition effect. Only repairs, which require the speaker to recognize an error, interrupt production, and formulate a correction, were significantly elevated. This specificity suggests that asynchronous context raises the criterion for tolerated error without globally impairing fluency. The monitoring system, under social-evaluative pressure, becomes more likely to act upon detected errors rather than to generate disfluencies indiscriminately.

4.3 Implications for Theories of Socially-Mediated Monitoring

These results extend the monitoring literature by demonstrating that social-contextual cues, independent of any manipulation of the physical feedback signal, are sufficient to modulate monitoring behavior. Prior work established that delayed auditory feedback engages right-lateralized monitoring networks (Kort et al., 2014) and that monitoring exhibits individual-difference correlates in these networks (Ozker et al., 2025). The present findings suggest that the monitoring system is sensitive to anticipated communicative consequences, not merely to degraded sensory input. This implies a top-down pathway through which social-cognitive representations of audience and communicative context can modulate the gain of the monitoring system,

consistent with broader predictive processing frameworks in which higher-level expectations tune lower-level perceptual and production processes.

4.4 Limitations

Several limitations qualify these findings. First, the sample size ($N = 21$), while adequate for detecting medium-to-large within-subjects effects, provided limited power for detecting small effects; the onset latency difference of approximately 40 ms ($d = 0.15$) may reflect a genuine but small effect that the current design was underpowered to detect. Second, the 5 s preview period, while ecologically reasonable for a picture description task, may have absorbed pre-articulatory condition effects; future studies could manipulate preview duration or use a paradigm with tighter temporal coupling between stimulus onset and speaking cue. Third, the manipulation of social context relied on instructional framing and experimenter behavior rather than a genuine communicative exchange; future studies could strengthen ecological validity by incorporating real-time audience feedback or manipulating audience characteristics parametrically. Fourth, the unequal and small English L1 subsample precludes firm conclusions about cross-language generalizability, though the directional consistency of the repair effect is encouraging.

5. Conclusion

This study provides the first experimental evidence that social-contextual cues alone, instantiated as asynchronous versus synchronous voice communication, can modulate the behavioral expression of speech self-monitoring. The effect is selective rather than global: asynchronous context elevates the proportion of self-repairs among disfluencies, driven by a specific increase in repair frequency, without significantly altering speech onset latency or the production of filled pauses, repetitions, or silent pauses. This dissociation supports a model in which social-evaluative pressure primarily upregulates post-articulatory monitoring, lowering the threshold for acting upon detected errors, while leaving pre-articulatory planning processes largely

unaffected. The selective nature of this effect refines Leveltian monitoring theory by situating the monitor within a social-cognitive architecture sensitive to the anticipated communicative future of the speech signal.

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